

Data-Driven Book Writing: Identifying Promising Genres for the Future of Books

Arafat Islam
BSc in Data Science
United International University
Dhaka, Bangladesh

Ahmed Hoque
BSc in Data Science
United International University
Dhaka, Bangladesh

Imran Hosain
BSc in Data Science
United International University
Dhaka, Bangladesh

Abstract—This project develops a practical data wrangling pipeline for transforming book records collected from Open Library into a structured dataset and using the resulting evidence to identify promising directions for future book writing. The workflow combines API-based data collection, asynchronous enrichment, data cleaning, missing-value handling, feature engineering, and visualization. The initial dataset contained 2,081 records and 17 columns, with substantial missingness in ratings, reading-shelf counts, page counts, and author information. After cleaning, duplicate consolidation, date standardization, validation of publication years, controlled imputation, and feature transformation, the project produced 2,075 book records with 33 analytical fields. The final dataset supports analysis of genre distribution, publication eras, book length, reader engagement, ebook availability, author information, and the relationship between edition count and reader ratings, with a final decision-oriented objective: if an author wants to write a new book, which genre appears most promising based on the observed data? The results show Romance as the largest genre with 281 books and the strongest observed reader-interest signal, while Young Adult has the highest mean rating among enriched records but only 10 books with rating data. Eight of the ten most-wanted books in the enriched subset are classified as Romance in the cleaned dataset. These findings suggest Romance is the strongest evidence-based genre choice for a new author within this dataset, while Young Adult represents a potentially promising quality-oriented direction that requires more data before a strong recommendation can be made. The project also explicitly separates observed values from imputed values and avoids fabricating missing engagement or rating data.

Index Terms—data wrangling, Open Library, data cleaning, feature engineering, API, book dataset, data visualization

I. INTRODUCTION

Books contain rich information about authors, genres, publication history, editions, reader interest, and digital availability. However, data obtained from public APIs is rarely analysis-ready. The same concept may appear in different formats, records may be duplicated under different categories, dates may use inconsistent representations, and many fields may be unavailable for a large portion of the dataset.

This project addresses that practical data-wrangling problem using book records from Open Library. Rather than treating the raw API response as a final dataset, the workflow separates collection, enrichment, cleaning, transformation, and visualization into reproducible stages. The main goal is to create a reliable analytical dataset while preserving the distinction be-

tween information that was actually observed and information that is unavailable.

The project is particularly focused on three questions: (1) how raw book metadata can be cleaned and standardized, (2) how additional analytical variables can be derived from the cleaned records, and (3) what useful patterns can be observed after the wrangling process.

II. PROJECT OBJECTIVES

The project was designed around the following objectives:

- Collect a book dataset through the Open Library API.
- Enrich book records with ratings, reading-shelf information, edition data, and author details.
- Detect and correct inconsistent formatting, invalid values, and duplicate records.
- Standardize genres and author date fields for reliable analysis.
- Handle missing values without inventing unsupported information.
- Create derived variables that make the dataset more useful for exploratory analysis.
- Generate visualizations that communicate the structure and major patterns of the cleaned data.
- Translate the observed reader-interest, rating, genre-size, and digital-availability patterns into a practical recommendation for future book writing.

III. DATA COLLECTION AND ENRICHMENT

The initial collection process used the Open Library Subjects API with the thriller subject endpoint and requested up to 40 records per request. The collection code extracted title, author, first publication year, edition count, ebook availability, genre-related information, and the Open Library work key. The resulting records were stored as a CSV file for subsequent processing.

The enrichment stage used asynchronous HTTP requests through Python's `aiohttp` library. For each work, the pipeline attempted to retrieve ratings, bookshelves, editions, and author details. The enrichment process used up to 25 concurrent connections, retry handling for HTTP 429 responses, timeout handling, a progress file, and an author cache. These mechanisms were included to reduce repeated requests and

make the process recoverable if execution stopped before all records were enriched.

The raw dataset contained 2,081 rows and 17 columns. Enrichment was incomplete at the time of the final wrangling run: useful rating, shelf, and author enrichment was available for 419 of 2,075 final records, or approximately 20%. The project therefore did not replace unavailable engagement values with artificial averages. Instead, an `Is_Enriched` flag was introduced and analyses involving ratings or reader engagement were restricted to the enriched subset.

IV. DATA WRANGLING METHODOLOGY

The complete pipeline can be viewed as a sequence of transformations from raw records to analysis-ready records, as illustrated conceptually below:

Raw Data → Cleaning → Validation
 → Missing Value Handling (1)
 → Feature Engineering → Visualization.

A. Formatting and Author Cleaning

Text fields such as titles, authors, and genre details were stripped of unnecessary whitespace and repeated spacing. Author strings were also cleaned to remove repeated names and prefixes such as “Adapted by”, “Edited by”, “Translated by”, and “Illustrated by”. Two useful fields were then derived: `Primary_Author`, representing the first author, and `Author_Count`, representing the number of authors recorded for the book. The process changed 31 author strings, while 220 records were identified as multi-author books.

B. Genre Standardization

The raw categorical representation was standardized into readable labels. For example, `science_fiction` became “Sci-Fi”, `young_adult` became “Young Adult”, and `historical_fiction` became “Historical Fiction”. Genre-detail subjects were also cleaned by removing repeated entries and selected Open Library metadata noise such as “open library staff picks” and “accessible book”.

C. Date Standardization

Author birth and death dates appeared in multiple formats. The pipeline converted these representations into ISO-style dates (YYYY-MM-DD) where sufficient information was available and also generated separate year fields. The raw birth-date field contained 37 identifiable formatting patterns, while the death-date field contained 30 patterns. Standardizing these values made it possible to perform chronological validation and calculate author age at publication.

D. Invalid and Implausible Values

Publication years outside the range 1450–2026 were treated as invalid. The values 1800 and 1900 were also treated as century-level placeholders rather than exact publication years. In total, 2 out-of-range years and 76 placeholder years were removed.

A second validation rule checked whether a book’s publication year occurred before the author’s birth year plus ten years. This identified 10 implausible publication records. The project also removed 36 cases where Open Library represented zero ratings as an average rating of zero; these values were interpreted as missing ratings rather than genuine zero scores. Page counts below 30 were treated as implausible and 3 such values were removed. No death-before-birth cases were found in the final validation run.

E. Duplicate Handling

Duplicate detection used normalized title and primary-author combinations rather than relying only on Open Library work keys. This was necessary because the same book could occur more than once under different genre classifications. Six duplicate rows were removed after selecting the most complete record and preserving the combined genre information in `All_Genres`. Records with the same title but different authors were retained because they may represent different works.

F. Missing Values

The project deliberately avoided broad statistical filling of missing ratings and engagement data. Such filling would create values that were not observed in the source. For enriched records, missing shelf counts were interpreted as zero where appropriate. Author biographies that were unavailable were marked as “Not available”.

Page counts required a more controlled strategy. For enriched records with missing page counts, the median page count of the corresponding genre was used only when at least five valid page counts existed for that genre. The affected records were marked with `Page_Count_Imputed`, allowing later analysis to distinguish measured and imputed page counts. A total of 97 enriched records received this controlled page-count imputation.

G. Feature Engineering

The final dataset expanded from 17 to 33 columns. New analytical variables included publication decade, historical era, book age, total reader count, completion rate, length category, popularity tier, author age at publication, author status, primary author, and author count.

The completion rate was calculated as:

$$\text{Completion Rate} = \frac{\text{Already Read}}{\text{Already Read} + \text{Currently Reading}}. \quad (2)$$

Length categories were defined as Short (<200 pages), Medium (200–350 pages), Long (350–500 pages), and Epic (500+ pages). Edition count was used to create popularity tiers ranging from Niche to Classic/Bestseller.

V. FINAL DATASET

After the wrangling process, the dataset contained 2,075 rows and 33 columns. Table I summarizes the main stages of the transformation.

TABLE I
DATASET TRANSFORMATION SUMMARY

Stage	Result
Raw records	2,081
Raw columns	17
Duplicate rows removed	6
Final records	2,075
Final columns	33
Enriched records	419
Enriched proportion	20%
Page counts imputed	97

Important fields in the final dataset include title, author, standardized genre, publication year, edition count, ebook availability, average rating, rating count, page count, reader-shelf counts, author dates, author biography, enrichment status, and the original Open Library key. The dataset is therefore suitable for both descriptive analysis and further machine-learning-oriented preprocessing.

VI. VISUALIZATION AND EXPLORATORY ANALYSIS

The project generated ten visualizations. They were designed to answer different questions rather than repeat the same information.

A. Genre Structure

Figure 1 shows the number of books by genre. Romance was the largest genre with 281 books, while Fantasy was the smallest with 228 books. The corresponding genre-distribution chart provides a proportional view of the same structure.

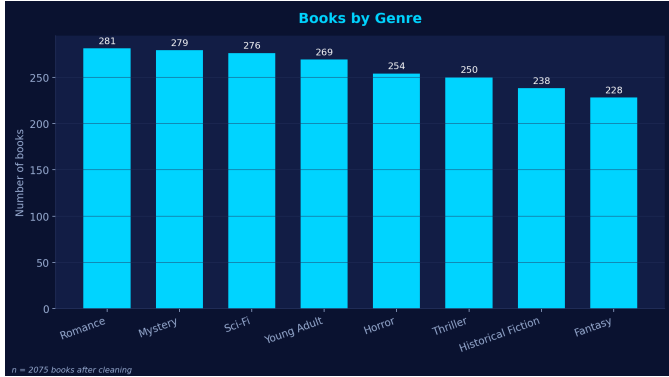


Fig. 1. Number of books by standardized genre.

B. Publication History

The publication-year analysis shows a median publication year of 1979, while the 2000s were the busiest decade with 308 books. The decade chart excludes records with missing or invalid publication years and treats the 2020s as a partial decade.

The era-level genre visualization further shows how the composition of the dataset changes across historical periods. This provides a more useful perspective than simply counting genres because it compares the relative contribution of each genre within an era.

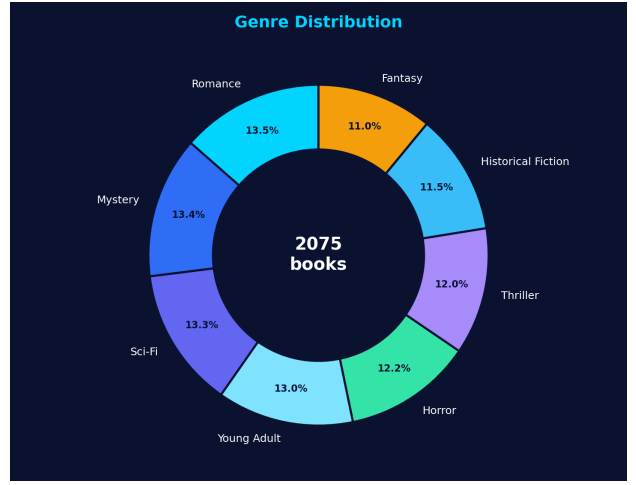


Fig. 2. Proportional genre distribution of the final dataset.

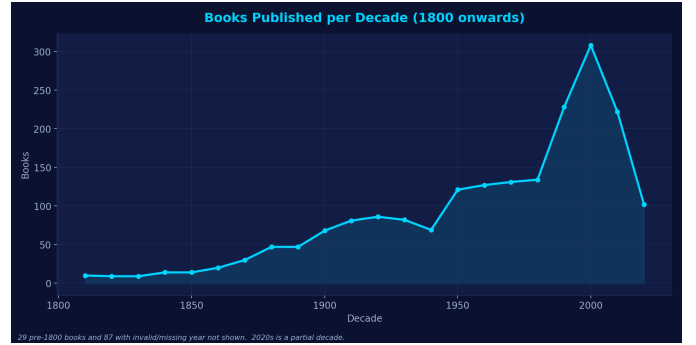


Fig. 3. Books published per decade from 1800 onward.

C. Ratings and Reader Interest

Rating analysis was restricted to books with available enrichment data. Young Adult had the highest mean rating at 4.34, while Fantasy had the lowest at 3.80. These results should be interpreted with the enrichment limitation in mind because they do not represent all 2,075 books.

The most wanted book in the enriched subset was *Fifty Shades of Grey*, with 6,030 want-to-read records. Figure 6 compares the top ten books by want-to-read counts and also

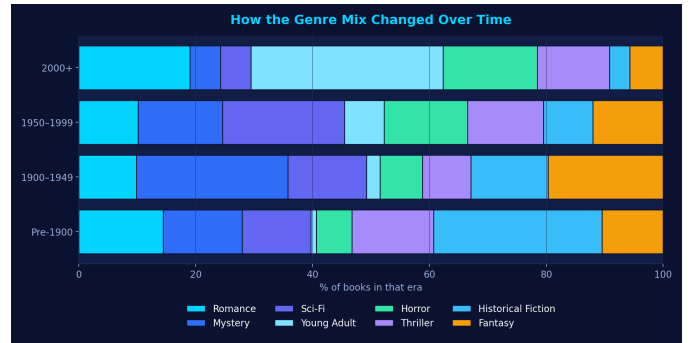


Fig. 4. Genre composition across publication eras.

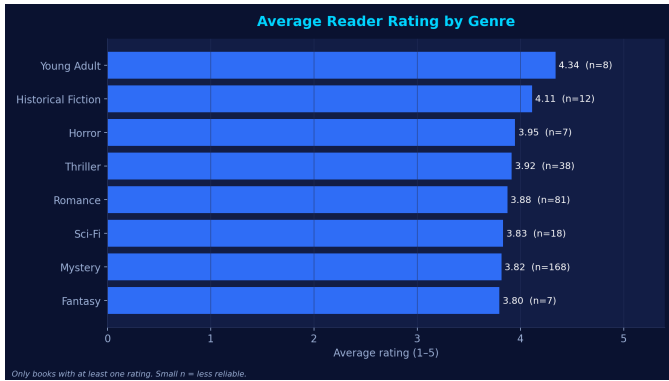


Fig. 5. Average reader rating by genre using enriched records.

displays already-read counts.

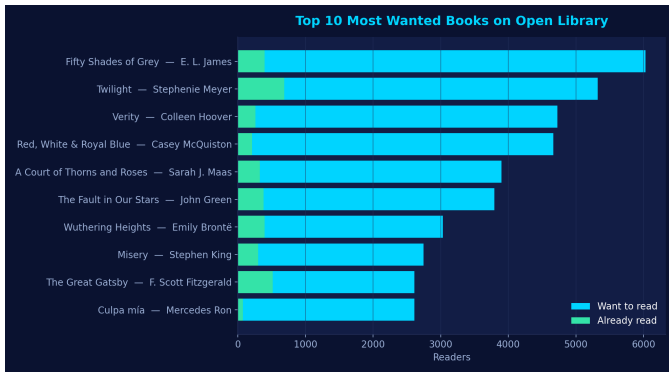


Fig. 6. Top ten most wanted books in the enriched dataset.

D. Book Length

The median observed book length was 308 pages. Figure 7 shows the distribution of page counts while excluding imputed values, which prevents the median and shape of the distribution from being artificially influenced by the genre-based replacement process.

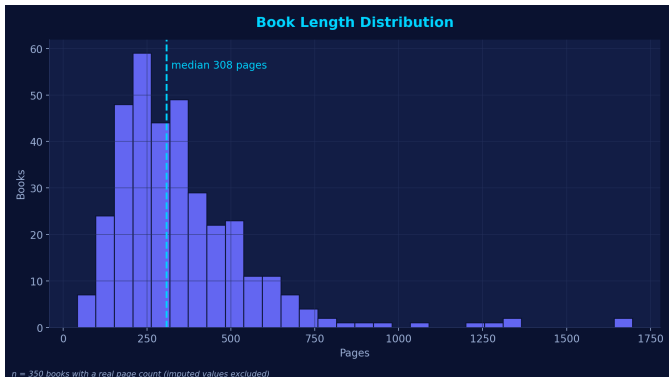


Fig. 7. Distribution of observed page counts; imputed values are excluded.

E. Popularity Versus Rating

Edition count was used as a practical indicator of bibliographic popularity. The project compared edition count with average reader rating using a log-scaled edition axis. The resulting correlation was $r = 0.09$, indicating a weak relationship. In other words, a book having many editions does not necessarily imply that it receives a substantially higher reader rating.

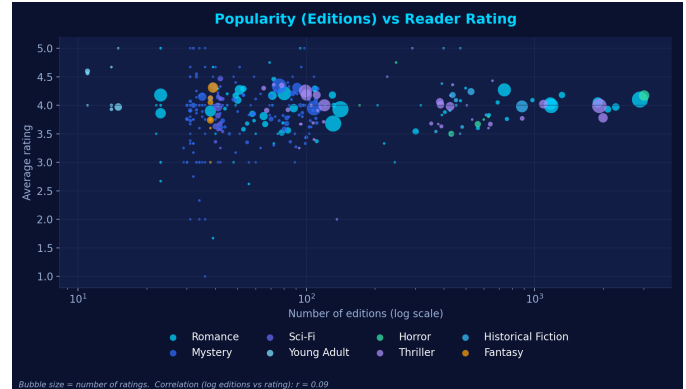


Fig. 8. Edition count versus average rating for enriched books. Bubble size represents rating count.

F. Impact of Wrangling

The project also generated a dedicated data-quality chart summarizing the number of records affected by different cleaning and standardization operations. This makes the wrangling stage measurable rather than presenting cleaning as an invisible preprocessing step.

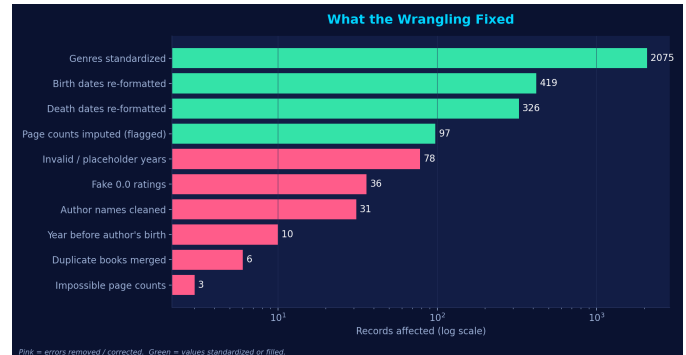


Fig. 9. Records affected by major data-quality fixes and transformations.

G. Ebook Availability

Ebook availability was analyzed across genres. Approximately 90% of the final dataset was marked as having full-text ebook availability according to the collected field, with Thriller reaching 100% in the generated genre-level analysis.

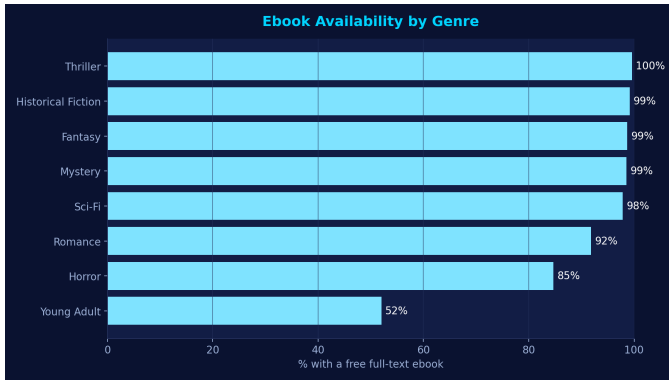


Fig. 10. Ebook availability by genre.

VII. DISCUSSION

The main value of the project is not simply the final charts but the data-quality decisions that make those charts more defensible. Several raw fields contained substantial missingness, so treating every blank as zero or filling every blank with a mean would have produced misleading conclusions. The explicit `Is_Enriched` flag provides a reproducible boundary between records with observed engagement information and records without it.

The duplicate-handling strategy also demonstrates why identifier-based deduplication is not always sufficient. Although Open Library work keys were unique, the same title and author could appear in multiple genre records. Combining these records preserved genre information while preventing double counting.

The weak editions-to-rating correlation is another useful example of why exploratory analysis should not rely on intuitive assumptions. Edition count captures a form of bibliographic reach, while average rating reflects reader evaluation. The observed $r = 0.09$ suggests that the two concepts are largely distinct within this dataset.

The publication-year analysis should also be interpreted carefully. Invalid and placeholder years were removed, and the 2020s are incomplete. Therefore, the decade counts describe the collected dataset rather than the global distribution of published books.

VIII. BOOK-WRITING OPPORTUNITY ANALYSIS

The final stage of the project moves from descriptive analysis to a practical decision question: *If a writer wants to create a new book, which genre should be considered based on the evidence in this dataset?* The answer is not treated as a prediction of the entire future publishing market. Instead, it is a data-driven recommendation limited to the observed Open Library sample.

Four signals were considered together: the number of books represented in each genre, the amount of observed reader interest, the average reader rating among enriched records, and digital availability. Reader interest was emphasized because the purpose of the question is not simply to identify which

genre exists most often, but which type of book appears capable of attracting readers. Rating was then used as a quality or satisfaction signal, while ebook availability was considered relevant to future digital accessibility.

TABLE II
EVIDENCE FOR FUTURE BOOK-WRITING DIRECTION

Genre	Books	Mean Rating	Observed Signal
Romance	281	3.87	Strongest overall reader-interest signal
Mystery	279	3.82	Large representation, moderate interest
Sci-Fi	276	3.85	Large representation, lower observed interest
Young Adult	269	4.36	Highest rating, but only 10 enriched ratings
Horror	254	4.00	Moderate rating and interest
Thriller	250	3.91	Strong reader interest in enriched subset
Historical Fiction	238	4.13	High rating, limited enrichment
Fantasy	228	3.84	Moderate rating and interest

A. Recommended Direction: Romance

Among the available evidence, **Romance is the strongest choice for a new book** if the author's primary objective is to maximize the probability of reader interest within the observed dataset. Romance is the largest genre with 281 books, and its enriched records contain the highest total want-to-read count among the genres with substantial engagement data. More importantly, the top-ten most-wanted list is dominated by Romance: *Fifty Shades of Grey*, *Twilight*, *Verity*, *Red, White & Royal Blue*, *A Court of Thorns and Roses*, *The Fault in Our Stars*, and other high-interest titles are classified as Romance in the cleaned dataset. This concentration provides a stronger demand signal than genre size alone.

The recommendation does not mean that every Romance book will succeed. Average rating for Romance is approximately 3.87, below Young Adult and Historical Fiction in this sample. Therefore, the evidence supports Romance primarily as a *reader-interest opportunity*, not as proof of superior literary quality.

B. A Second Direction: Young Adult

Young Adult provides an interesting alternative. It has the highest mean rating in the enriched analysis, approximately 4.36, suggesting strong reader satisfaction among the books for which rating information was available. However, only 10 Young Adult books contained rating data in the enriched subset. Consequently, the rating result is promising but statistically fragile compared with the much larger Romance rating sample. A writer interested in Young Adult may therefore consider it a high-potential direction, but additional data should be collected before treating it as the primary recommendation.

C. What the Data Suggest for Future Books

The strongest practical interpretation is to prioritize a genre with demonstrated reader interest while designing the book around qualities associated with high reader satisfaction. Within this project, that leads to the following decision framework:

- 1) **Primary choice: Romance** because it combines the largest genre representation with the strongest observed reader-interest concentration.
- 2) **Alternative: Young Adult** because it has the highest observed average rating, although the available rating sample is small.
- 3) **Consider Thriller as a secondary option** because its enriched records show meaningful want-to-read activity and its ebook availability is very high.
- 4) **Do not use edition count alone as a success predictor.** The correlation between edition count and rating is weak ($r = 0.09$), so bibliographic reach and reader satisfaction should be treated as separate signals.
- 5) **Prefer digital accessibility.** The high ebook-availability rates across most genres indicate that future books can benefit from being designed and distributed for both physical and digital readers.

Therefore, if the question is simply “*What should I write?*”, the project recommends **a Romance-oriented book as the safest evidence-based choice within this dataset**. If the question is “*What direction may provide stronger reader satisfaction?*”, Young Adult deserves further investigation. A future version of the project should collect a larger and more balanced enriched sample before making a stronger market-level prediction.

IX. LIMITATIONS

The project has several limitations. First, the dataset is based on a specific Open Library API collection rather than a complete census of books. Second, the enrichment process was incomplete, leaving only 419 records with usable rating and shelf information at the time of wrangling. Third, genre is based on Open Library subject/category information and therefore reflects source metadata rather than an independently verified literary classification.

The page-count imputation is intentionally conservative, but it remains an estimate. Similarly, edition count is only a proxy for popularity and should not be interpreted as sales. The project explicitly avoids making sales-based conclusions because sales data was not available in the collected source.

X. CONCLUSION

This project demonstrates that data wrangling can be used not only to clean a dataset but also to support a practical decision. Starting from 2,081 raw records, the pipeline produced 2,075 cleaned records and expanded the analytical representation from 17 to 33 columns. The workflow addressed inconsistent formatting, genre normalization, mixed date formats, invalid values, duplicate records, missing values, controlled imputation, and feature engineering.

The exploratory analysis produced several decision-relevant findings. Romance was the largest genre with 281 books, and it showed the strongest concentration of reader-interest signals in the enriched data. Eight of the ten most-wanted books were classified as Romance, making it the clearest genre-level demand signal in this project. Young Adult had the highest mean rating at approximately 4.36, but this result was based on only 10 enriched rating records and therefore requires further validation. Thriller also showed strong reader-interest activity and nearly complete ebook availability.

Based on these findings, the project recommends **Romance as the most promising genre for an author who wants to write a new book using the evidence available in this dataset**. This should be interpreted as a data-supported opportunity rather than a guarantee of future commercial success. The analysis suggests that future book development should combine demonstrated reader interest with attention to reader satisfaction and digital accessibility rather than relying on a single metric.

The most important lesson is that a cleaned dataset can turn a broad question such as “Which book should I write?” into a structured decision problem. With a larger enriched dataset, additional variables such as publication growth, review sentiment, author popularity, regional readership, and time-based trends could make this recommendation more robust and more suitable for future forecasting.

ACKNOWLEDGMENT

The authors acknowledge the Open Library project for providing publicly accessible bibliographic data and the course environment that supported the development of this data-wrangling workflow.

REFERENCES

- [1] Open Library, “Open Library APIs,” Open Library. [Online]. Available: <https://openlibrary.org/developers/api>
- [2] The pandas development team, “pandas documentation.” [Online]. Available: <https://pandas.pydata.org/docs/>
- [3] AIOHTTP contributors, “aiohttp documentation.” [Online]. Available: <https://docs.aiohttp.org/>
- [4] J. D. Hunter, “Matplotlib: A 2D graphics environment,” *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.